

Module 7: Handwritten Learning Evidence

Section 7.2 – Best Approximation and Least Squares

Question 1 – Nicholson Exercise 5.6.4(a)

Question.

Find a least-squares approximating function of the form

$$f(x) = r_0x + r_1x^2 + r_2 2^x$$

for the data pairs

$$(-1, 1), \quad (0, 3), \quad (1, 1), \quad (2, 0).$$

Concepts / properties applied:

Theorem 5.6.1 (Best Approximation Theorem); Theorem 5.6.4 (general least-squares approximation); normal equations.

Method.

Use the three basis functions

$$x, \quad x^2, \quad 2^x$$

to form the matrix M , then solve the normal equations

$$(M^T M)z = M^T y.$$

Step-by-step working

Let

$$z = \begin{bmatrix} r_0 \\ r_1 \\ r_2 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 3 \\ 1 \\ 0 \end{bmatrix}.$$

Evaluating the basis functions at the given x -values gives

$$M = \begin{bmatrix} -1 & 1 & \frac{1}{2} \\ 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 4 & 4 \end{bmatrix}.$$

Hence,

$$(M^T M)z = M^T y \quad (\text{Theorem 5.6.1}).$$

First,

$$M^T = \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 4 \\ \frac{1}{2} & 1 & 2 & 4 \end{bmatrix}.$$

Therefore,

$$\begin{aligned}
M^T M &= \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 4 \\ \frac{1}{2} & 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 & \frac{1}{2} \\ 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 4 & 4 \end{bmatrix} \\
&= \begin{bmatrix} 6 & 8 & \frac{19}{2} \\ 8 & 19 & \frac{39}{2} \\ \frac{19}{2} & \frac{39}{2} & \frac{85}{4} \end{bmatrix} \quad (\text{Matrix multiplication}).
\end{aligned}$$

Also,

$$M^T y = \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 4 \\ \frac{1}{2} & 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ \frac{11}{2} \end{bmatrix} \quad (\text{Matrix multiplication}).$$

Thus,

$$\begin{bmatrix} 6 & 8 & \frac{19}{2} \\ 8 & 19 & \frac{39}{2} \\ \frac{19}{2} & \frac{39}{2} & \frac{85}{4} \end{bmatrix} \begin{bmatrix} r_0 \\ r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ \frac{11}{2} \end{bmatrix}.$$

Multiplying through by 4,

$$\begin{aligned}
24r_0 + 32r_1 + 38r_2 &= 0, \\
32r_0 + 76r_1 + 78r_2 &= 20, \quad (\text{Equivalent system}) \\
38r_0 + 78r_1 + 85r_2 &= 22.
\end{aligned}$$

Eliminating r_0 by combining the first two equations gives

$$100r_1 + 82r_2 = 60 \quad (\text{Gaussian elimination}).$$

Similarly, eliminating r_0 using the first and third equations gives

$$164r_1 + 149r_2 = 132 \quad (\text{Gaussian elimination}).$$

From the first equation,

$$r_1 = \frac{30 - 41r_2}{50}.$$

Substituting into the second equation,

$$164 \left(\frac{30 - 41r_2}{50} \right) + 149r_2 = 132,$$

so

$$r_2 = \frac{280}{121}.$$

Therefore,

$$r_1 = \frac{30 - 41 \left(\frac{280}{121} \right)}{50} = -\frac{157}{121}.$$

Finally, using

$$24r_0 + 32r_1 + 38r_2 = 0,$$

we obtain

$$r_0 = -\frac{234}{121}.$$

Verification

Using these coefficients,

$$Mz = \begin{bmatrix} \frac{217}{121} \\ \frac{123}{121} \\ \frac{169}{121} \\ \frac{24}{121} \end{bmatrix}.$$

Hence,

$$y - Mz = \begin{bmatrix} -\frac{96}{121} \\ \frac{240}{121} \\ -\frac{48}{121} \\ -\frac{24}{121} \end{bmatrix}.$$

Now,

$$M^T(y - Mz) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (\text{Normal-equation verification}).$$

Thus the residual satisfies the required orthogonality condition.

Final answer

$$f(x) = -\frac{234}{121}x - \frac{157}{121}x^2 + \frac{280}{121}2^x$$

Key Learning Point.

The least-squares method is not restricted to straight-line fitting. The same normal-equation framework applies to a linear combination of different basis functions, with the columns of M representing those functions.

Question 2 – Nicholson Exercise 5.6.11

Question.

Assume $n = m + 1$ in Theorem 5.6.3, so that M is square. If the x_i are distinct, show that M is invertible. Deduce that

$$z = M^{-1}y$$

and that the least-squares polynomial is the interpolating polynomial and actually passes through all the data points.

Concepts / properties applied:

Theorem 3.2.6 (interpolating polynomial); Theorem 5.6.1 (Best Approximation Theorem); Theorem 5.6.3 (least-squares polynomial); Vandermonde determinant; invertibility criterion.

Method.

Since $n = m + 1$, M is a square Vandermonde matrix. Show that its determinant is non-zero for distinct x_i , then simplify the normal equations.

Step-by-step working

The matrix is

$$M = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^m \\ 1 & x_2 & x_2^2 & \cdots & x_2^m \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^m \end{bmatrix}.$$

Since the x_i are distinct,

$$\det(M) = \prod_{1 \leq i < j \leq n} (x_j - x_i) \neq 0 \quad (\text{Vandermonde determinant}).$$

Therefore,

$$M \text{ is invertible} \quad (\text{Invertibility criterion}).$$

The normal equations are

$$(M^T M)z = M^T y \quad (\text{Theorem 5.6.1}).$$

Since M and M^T are invertible,

$$\begin{aligned} z &= (M^T M)^{-1} M^T y \\ &= M^{-1} (M^T)^{-1} M^T y \quad (\text{Inverse-matrix property}) \\ &= M^{-1} y. \end{aligned}$$

Thus,

$$\boxed{z = M^{-1} y}.$$

Verification

Multiplying by M ,

$$Mz = y \quad (\text{Matrix multiplication}).$$

But

$$Mz = \begin{bmatrix} p(x_1) \\ p(x_2) \\ \vdots \\ p(x_n) \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}.$$

Therefore,

$$p(x_i) = y_i, \quad i = 1, \dots, n \quad (\text{Interpolation condition}).$$

Hence,

$$\|y - Mz\| = 0.$$

The least-squares polynomial therefore passes exactly through every data point.

Final answer

$$\det(M) \neq 0 \implies M \text{ is invertible}$$

$$z = M^{-1}y$$

and

$$p(x_i) = y_i \quad \text{for every } i$$

so the least-squares polynomial is exactly the interpolating polynomial.

Key Learning Point.

With exactly $m + 1$ data points and distinct x_i -values, the least-squares problem becomes exact interpolation: the minimum possible error is zero.

Question 3 – Nicholson Exercise 5.6.12

Question.

Let A be any $m \times n$ matrix and define

$$K = \{x : A^T A x = 0\}.$$

Let b be an m -column. Show that, if z is an n -column such that $\|b - Az\|$ is minimal, then all such vectors have the form

$$z + x$$

for some $x \in K$.

Concepts / properties applied:

Theorem 5.6.1 (Best Approximation Theorem); normal equations; linearity of matrix multiplication; definition of K .

Method.

Take two vectors that both produce the minimum residual. Show that their difference belongs to K , then prove the converse by showing that adding any vector from K preserves the normal equations.

Step-by-step working

Since z gives the minimum value of

$$\|b - Az\|,$$

it satisfies

$$A^T Az = A^T b \quad (\text{Theorem 5.6.1}).$$

Now let y be another vector giving the same minimum. Then

$$A^T Ay = A^T b \quad (\text{Normal-equation characterisation}).$$

Therefore,

$$A^T Ay = A^T Az.$$

Subtracting gives

$$A^T A(y - z) = 0 \quad (\text{Linearity}).$$

Let

$$x = y - z.$$

Then

$$A^T Ax = 0,$$

so

$$x \in K \quad (\text{Definition of } K).$$

Therefore,

$$y = z + x, \quad x \in K.$$

Converse

Now take any $x \in K$. Then

$$A^T Ax = 0 \quad (\text{Definition of } K).$$

Since z satisfies the normal equations,

$$A^T Az = A^T b.$$

Therefore,

$$\begin{aligned} A^T A(z + x) &= A^T Az + A^T Ax \\ &= A^T b + 0 && \text{(Linearity)} \\ &= A^T b. \end{aligned}$$

Thus $z + x$ also satisfies the normal equations and is therefore also a best approximation.

(Theorem 5.6.1)

Verification

We have established both directions:

$$y \text{ is a minimiser} \implies y = z + x, \quad x \in K,$$

and

$$x \in K \implies z + x \text{ is a minimiser.}$$

Therefore, the complete set of best-approximation vectors is

$$z + K \quad \text{(Solution-set structure).}$$

Final answer

$$\boxed{\{y : \|b - Ay\| \text{ is minimal}\} = z + K}$$

where

$$\boxed{K = \{x : A^T Ax = 0\}.$$

Equivalently,

$$\boxed{y = z + x, \quad x \in K.}$$

Key Learning Point.

The normal equations describe the complete family of minimisers when uniqueness is not guaranteed. Any two best-approximation vectors differ by a vector in K , linking least-squares approximation with null-space structure.

Section 7.3 – Examples and Basic Properties of Abstract Vector Spaces

Question 1 – Nicholson Exercise 6.1.4

Question.

Let V be the set of ordered pairs $(x, y) \in \mathbb{R}^2$, with the operations

$$(x, y) + (x_1, y_1) = (x + x_1, y + y_1 + 1)$$

and

$$a(x, y) = (ax, ay + a - 1).$$

Show that V is a vector space and determine its zero vector.

Concepts / properties applied:

A1–A5; S1–S5; closure; zero vector; additive inverse; field properties.

Method.

Because the operations are non-standard, verify the vector-space axioms directly using the operations given.

Step-by-step working

A1 – Closure under addition

For $(x, y), (x_1, y_1) \in V$,

$$(x, y) + (x_1, y_1) = (x + x_1, y + y_1 + 1) \in \mathbb{R}^2. \quad (\text{A1})$$

Hence the sum remains in V .

A2 – Commutativity

$$\begin{aligned} (x, y) + (x_1, y_1) &= (x + x_1, y + y_1 + 1) \\ &= (x_1 + x, y_1 + y + 1) \quad (\text{A2}) \\ &= (x_1, y_1) + (x, y). \end{aligned}$$

A3 – Associativity

$$\begin{aligned} ((x, y) + (x_1, y_1)) + (x_2, y_2) &= (x + x_1, y + y_1 + 1) + (x_2, y_2) \\ &= (x + x_1 + x_2, y + y_1 + y_2 + 2), \end{aligned}$$

while

$$\begin{aligned} (x, y) + ((x_1, y_1) + (x_2, y_2)) &= (x, y) + (x_1 + x_2, y_1 + y_2 + 1) \\ &= (x + x_1 + x_2, y + y_1 + y_2 + 2). \end{aligned}$$

Therefore,

$$\boxed{\left((x, y) + (x_1, y_1)\right) + (x_2, y_2) = (x, y) + \left((x_1, y_1) + (x_2, y_2)\right)}. \quad (\text{A3})$$

A4 – Zero vector

Let the zero vector be (p, q) . Then

$$(x, y) + (p, q) = (x, y).$$

Hence,

$$(x + p, y + q + 1) = (x, y),$$

so

$$p = 0, \quad q = -1.$$

Thus,

$$\boxed{\mathbf{0} = (0, -1)}. \quad (\text{A4})$$

A direct check gives

$$(x, y) + (0, -1) = (x, y - 1 + 1) = (x, y).$$

A5 – Additive inverse

Let

$$-(x, y) = (r, s).$$

Then

$$(x, y) + (r, s) = (0, -1),$$

so

$$x + r = 0, \quad y + s + 1 = -1.$$

Therefore,

$$r = -x, \quad s = -y - 2.$$

Hence,

$$\boxed{-(x, y) = (-x, -y - 2)}. \quad (\text{A5})$$

Indeed,

$$(x, y) + (-x, -y - 2) = (0, -1).$$

S1 – Closure under scalar multiplication

For $a \in \mathbb{R}$,

$$a(x, y) = (ax, ay + a - 1) \in \mathbb{R}^2. \quad (\text{S1})$$

Thus scalar multiplication remains in V .

S2 – Distributivity over vector addition

$$\begin{aligned} a((x, y) + (x_1, y_1)) &= a(x + x_1, y + y_1 + 1) \\ &= (a(x + x_1), a(y + y_1 + 1) + a - 1) \\ &= (ax + ax_1, ay + ay_1 + 2a - 1). \end{aligned}$$

On the other hand,

$$\begin{aligned} a(x, y) + a(x_1, y_1) &= (ax, ay + a - 1) + (ax_1, ay_1 + a - 1) \\ &= (ax + ax_1, ay + ay_1 + 2a - 1). \end{aligned}$$

Therefore,

$$\boxed{a(u + v) = au + av}. \quad (\text{S2})$$

S3 – Distributivity over scalar addition

$$(a + b)(x, y) = ((a + b)x, (a + b)y + a + b - 1).$$

Also,

$$\begin{aligned} a(x, y) + b(x, y) &= (ax, ay + a - 1) + (bx, by + b - 1) \\ &= ((a + b)x, (a + b)y + a + b - 1). \end{aligned}$$

Hence,

$$\boxed{(a + b)v = av + bv}. \quad (\text{S3})$$

S4 – Compatibility of scalar multiplication

$$\begin{aligned} a(b(x, y)) &= a(bx, by + b - 1) \\ &= (abx, a(by + b - 1) + a - 1) \\ &= (abx, aby + ab - 1). \end{aligned}$$

But

$$(ab)(x, y) = (abx, aby + ab - 1).$$

Therefore,

$$\boxed{a(bv) = (ab)v}. \quad (\text{S4})$$

S5 – Scalar identity

$$1(x, y) = (x, y + 1 - 1) = (x, y). \quad (\text{S5})$$

Thus,

$$1v = v.$$

Verification

All ten vector-space properties have been checked under the specified operations:

$$\text{A1–A5 and S1–S5 all hold in } V.$$

In particular, the zero vector and additive inverse satisfy their required identities.

Final answer

$$V \text{ is a vector space over } \mathbb{R}$$

with

$$\mathbf{0} = (0, -1).$$

Also,

$$-(x, y) = (-x, -y - 2).$$

Key Learning Point.

The vector-space axioms depend on the structure of the operations rather than on their familiar appearance. The zero vector and additive inverse must therefore be determined from the operations actually defined.

Question 2 – Nicholson Exercise 6.1.2(g)

Question.

Determine whether the set V of all 2×2 matrices with zero determinant, using the usual matrix operations, is a vector space. If not, give a reason.

Concepts / properties applied:

Closure under addition; determinant; counterexample.

Method.

To disprove that V is a vector space, it is sufficient to find two matrices in V whose sum is not in V .

Step-by-step working

Choose

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$\det(A) = 1(0) - 0(0) = 0,$$

and

$$\det(B) = 0(1) - 0(0) = 0.$$

Therefore,

$$A, B \in V. \quad (\text{Given condition})$$

Now add the two matrices:

$$A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2.$$

But

$$\det(A + B) = \det(I_2) = 1 \neq 0. \quad (\text{Determinant})$$

Therefore,

$$A + B \notin V.$$

Hence V is not closed under addition.

Verification

We have explicitly found

$$A, B \in V$$

but

$$A + B \notin V.$$

Therefore the closure requirement fails.

Final answer

The set of 2×2 matrices with determinant 0 is not a vector space.

More specifically,

it is not closed under addition.

Key Learning Point.

A subset of a vector space is not automatically a vector space. The zero-determinant condition is not preserved by addition, so one valid counterexample is enough to rule out the vector-space structure.

Question 3 – Nicholson Exercise 6.1.15(a)

Question.

Prove that, for a vector v and scalar a ,

$$av = 0$$

implies

$$a = 0 \quad \text{or} \quad v = 0.$$

Concepts / properties applied:

Vector-space axioms; scalar inverse; scalar associativity; cancellation; zero-vector property.

Method.

Start from $av = 0$. The case $a = 0$ is immediate. For $a \neq 0$, multiply by a^{-1} and simplify using the vector-space properties.

Step-by-step working

Suppose

$$av = 0.$$

If

$$a = 0,$$

then the required conclusion is immediate.

Now consider the remaining case,

$$a \neq 0.$$

Therefore a^{-1} exists. Multiplying both sides by a^{-1} ,

$$a^{-1}(av) = a^{-1}0. \quad (\text{Scalar multiplication})$$

Using scalar associativity,

$$(a^{-1}a)v = a^{-1}0. \quad (\text{S4})$$

Since

$$a^{-1}a = 1,$$

we obtain

$$v = a^{-1}0. \quad (\text{S5})$$

Now, because

$$0 = 0 + 0,$$

we have

$$\begin{aligned} a^{-1}0 &= a^{-1}(0 + 0) \\ &= a^{-1}0 + a^{-1}0. \end{aligned} \quad (\text{S2})$$

Adding the additive inverse of $a^{-1}0$ to both sides gives

$$a^{-1}0 = 0. \quad (\text{A5, cancellation})$$

Hence,

$$v = 0.$$

Thus,

$$a \neq 0 \implies v = 0.$$

Combining this with the case $a = 0$,

$$\boxed{av = 0 \implies a = 0 \text{ or } v = 0}.$$

Verification

The two possible cases are

$$a = 0,$$

or

$$a \neq 0 \implies v = 0.$$

Therefore the required implication holds for every scalar a .

Final answer

$$\boxed{av = 0 \implies a = 0 \text{ or } v = 0}$$

Key Learning Point.

This property follows from the vector-space axioms rather than from the usual coordinate interpretation of vectors. The non-zero scalar case is resolved by its multiplicative inverse, together with the scalar multiplication axioms.

Question 1

The system of equations,

$$\begin{aligned} -3x - 2y &= 3 \\ x - y &= -6 \\ -2x - 2y &= -5 \end{aligned}$$

where $x, y \in \mathbb{R}$, has no solution.

Find the vector $\mathbf{z} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ that best approximates a solution.

$$\mathbf{z} = \begin{pmatrix} \boxed{-3} \\ \boxed{4.111} \end{pmatrix} \checkmark$$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Your answer is correct. You were awarded **1** mark. 

You scored **1** mark for this part.

Score: 1/1 

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Question 1

The system of equations,

$$\begin{aligned} -6x + 4y &= 2 \\ 5x - y &= 4 \\ 4x - y &= 4 \end{aligned}$$

where $x, y \in \mathbb{R}$, has no solution.

Find the vector $\mathbf{z} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ that best approximates a solution.

$$\mathbf{z} = \begin{pmatrix} \boxed{1.455} \\ \boxed{2.667} \end{pmatrix} \checkmark$$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Your answer is correct. You were awarded **1** mark. 

You scored **1** mark for this part.

Score: 1/1 

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Question 1

The system of equations,

$$\begin{aligned}x - 4y &= -3 \\ 6x - 5y &= -6 \\ x + 2y &= 5\end{aligned}$$

where $x, y \in \mathbb{R}$, has no solution.

Find the vector $\mathbf{z} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ that best approximates a solution.

$$\mathbf{z} = \begin{pmatrix} \boxed{0.195} \\ \boxed{1.294} \end{pmatrix} \quad \checkmark$$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Your answer is correct. You were awarded **1** mark. 

You scored **1** mark for this part.

Score: 1/1 

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Question 1

Find the least squares line of best fit $y = a + bx$ for the data points:

$$\{(-4, -6), (-6, 0), (4, 4), (-5, -1)\}$$

$a =$ ✓

$b =$ ✓

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Gap 0

Your answer is correct. You were awarded **1** mark. ✓

Gap 1

Your answer is correct. You were awarded **1** mark. ✓

You scored **2** marks for this part.

Score: 2/2 ✓

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Question 1

Find the least squares line of best fit $y = a + bx$ for the data points:

$$\{(5, 2), (-5, -2), (-4, 3), (-2, 3)\}$$

$$a = \boxed{1.77} \quad \checkmark$$

$$b = \boxed{0.18} \quad \checkmark$$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Gap 0

Your answer is correct. You were awarded **1** mark. 

Gap 1

Your answer is correct. You were awarded **1** mark. 

You scored **2** marks for this part.

Score: 2/2 

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Question 1

Find a basis and calculate the dimension for the following subspace of 2×2 matrices (with standard addition and multiplication by a scalar operations)

$$\text{span} \left\{ \begin{pmatrix} 3 & 4 \\ 6 & 7 \end{pmatrix}, \begin{pmatrix} 2 & 3 \\ 2 & 2 \end{pmatrix}, \begin{pmatrix} -3 & -6 \\ 0 & -6 \end{pmatrix}, \begin{pmatrix} 3 & 9 \\ -7 & 10 \end{pmatrix} \right\}$$

Enter up to 4 basis matrices. For any matrices not required in the basis, enter

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

$$\begin{pmatrix} \boxed{3} & \boxed{4} \\ \boxed{6} & \boxed{7} \end{pmatrix}, \begin{pmatrix} \boxed{2} & \boxed{3} \\ \boxed{2} & \boxed{2} \end{pmatrix}, \begin{pmatrix} \boxed{-3} & \boxed{-6} \\ \boxed{0} & \boxed{-6} \end{pmatrix}, \begin{pmatrix} \boxed{3} & \boxed{9} \\ \boxed{-7} & \boxed{10} \end{pmatrix}$$

What is the dimension of the subspace?

Correct. Your basis matrices are independent and span the entire subspace. 
You were awarded **5** marks.

You scored **5** marks for this part.

Score: 5/5 

Question 1

The vector space spanned by the following set of polynomials is a subspace of polynomials of degree $n \leq 5$.

$$\{3x^3 - 2x^2 - 2x^4 + 2x^5, \quad 2x^2 - 3x^3 + 4x^4 + 2x^5, \quad 3x^2 - 6x^3 - 2x^4 - 1\}$$

Identify a basis of the subspace. Enter the coefficients in a matrix such that

Rows: Columns:

$$\begin{pmatrix} \begin{matrix} -2 & 3 & -2 & 2 \end{matrix} \\ \begin{matrix} 2 & -3 & 4 & 2 \end{matrix} \\ \begin{matrix} 3 & -6 & -2 & -10 \end{matrix} \end{pmatrix} \begin{pmatrix} x^2 \\ x^3 \\ x^4 \\ x^5 \end{pmatrix} = \begin{pmatrix} \text{basis polynomials} \end{pmatrix}$$

(reduce the size of the matrix if the dimension is less than 4)

What is the dimension of the subspace?

Correct. Your basis polynomials are independent and span the entire subspace. ✓
You were awarded **2** marks.

You scored **2** marks for this part.

Score: 2/2 ✓

Consider the set of ordered pairs defined on the Cartesian product

$V = \{0, 1, 2\} \times \{0, 1, 2\}$, i.e. pairs of the form $\mathbf{x} = (x_1, x_2) \in V$ with $x_1 \in \{0, 1, 2\}$ and $x_2 \in \{0, 1, 2\}$. We can also refer to the pairs as 2-dimensional vectors.

We define scalar multiplication of elements in V such that for a scalar $a \in \{0, 1, 2\}$, we have $a \otimes \mathbf{x} = (a \otimes x_1, a \otimes x_2)$. Furthermore, vector addition between elements in V is defined such that $\mathbf{x} \oplus \mathbf{y} = (x_1 \oplus y_1, x_2 \oplus y_2)$. The addition ($\oplus$) and multiplication ($\otimes$) operations for scalars are defined according to the following tables.

$\oplus$	0	1	2	$\otimes$	0	1	2
0	0	1	2	0	0	0	0
1	1	2	0	1	0	1	2
2	2	0	1	2	0	2	1

The set of vectors in V along with $\otimes$ and $\oplus$ defines a vector space. The following will ask you to provide examples for each property (see p328 of the text). Note that the examples you'll provide don't **prove** that the properties are satisfied, but rather allow you to get an intuitive feel for how the properties work. The standard symbol '+' will be used to mean $\oplus$ in each of the axioms.

a)

First, give three distinct examples for vectors in V . Do not include $(0, 0)$.

$$\mathbf{u} = \left(\boxed{1} \quad \boxed{0} \right)$$

$$\mathbf{v} = \left(\boxed{0} \quad \boxed{1} \right)$$

$$\mathbf{w} = \left(\boxed{1} \quad \boxed{1} \right)$$

Proceed to the next part. You were awarded **3** marks. ✓

You scored **3** marks for this part.

Score: 3/3 ✓

b)

Axioms for Vector Addition

For each of the following, fill the gaps for the examples. Note that $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ should correspond with the vectors you provided in part a (write these down on paper to save yourself scrolling up and down).

A1. If $\mathbf{u}$ and $\mathbf{v}$ are in V , then $\mathbf{u} + \mathbf{v}$ is in V .

Example: $\mathbf{u}$ and $\mathbf{v}$ are in V , the vector $\mathbf{u} + \mathbf{v} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$ ✓ is also in V .

A2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u}, \mathbf{v}$ in V .

Example: $\mathbf{u} + \mathbf{v} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$ ✓, $\mathbf{v} + \mathbf{u} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$ ✓

A3. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ for all $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ in V .

Example: $\mathbf{v} + \mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{2} \end{pmatrix}$ ✓, $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = \begin{pmatrix} \boxed{2} & \boxed{2} \end{pmatrix}$ ✓

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix} \checkmark, \quad (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \begin{pmatrix} \boxed{2} & \boxed{2} \end{pmatrix} \checkmark$$

A4. An element $\mathbf{0}$ in V exists such that $\mathbf{v} + \mathbf{0} = \mathbf{v} = \mathbf{0} + \mathbf{v}$ for every $\mathbf{v}$ in V .

Example: The zero element is $\mathbf{0} = \begin{pmatrix} \boxed{0} & \boxed{0} \end{pmatrix} \checkmark$.

A5. For each $\mathbf{v}$ in V , an element $-\mathbf{v}$ in V exists such that $-\mathbf{v} + \mathbf{v} = \mathbf{0}$ and $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.

Example: $-\mathbf{u} = \begin{pmatrix} \boxed{2} & \boxed{0} \end{pmatrix} \checkmark$, and $-\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix} \checkmark$.

Gap 0

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark.

Gap 1

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark.

Gap 2

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark.

Gap 3

Your answer is correct. You were awarded 1 mark.

Gap 4

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded **1** mark. ✓

Gap 5

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded **1** mark. ✓

Gap 6

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded **1** mark. ✓

Gap 7

Your answer is correct. You were awarded **1** mark. ✓

Gap 8

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded **1** mark. ✓

Gap 9

Your answer is correct. You were awarded **1** mark. ✓

You scored **10** marks for this part.

Score: 10/10 ✓

c)

Now enter 2 values, a , b , with both taken from $\{0, 1, 2\}$. They do not have to be different but ensure that $a \neq 0$.

$a =$

$b =$

Proceed to the next part. You were awarded **2** marks. ✓

You scored **2** marks for this part.

d)

Axioms for Scalar Multiplication

As with the addition axioms, fill out the examples below using your values of a and b , as well as the same $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ from part a. (write them all down to save you scrolling up)

Note that $a + b$ is defined the same as for $a \oplus b$ given in the table, and that in contrast to the axioms of page 328 of the text, here we assume a, b are in $\{0, 1, 2\}$ rather than in $\mathbb{R}$ (this structure is called a 'Field').

S1. If $\mathbf{v}$ is in V , then $a\mathbf{v}$ is in V for all a in $\{0, 1, 2\}$.

Example: $a\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix} \checkmark$, which is in V .

S2. $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$ for all $\mathbf{v}$ and $\mathbf{w}$ in V and all a in $\{0, 1, 2\}$.

Example: $\mathbf{v} + \mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{2} \end{pmatrix} \checkmark$, $a(\mathbf{v} + \mathbf{w}) = \begin{pmatrix} \boxed{1} & \boxed{2} \end{pmatrix} \checkmark$.

$a\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix} \checkmark$, $a\mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix} \checkmark$, $a\mathbf{v} + a\mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{2} \end{pmatrix} \checkmark$.

S3. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$ for all $\mathbf{v}$ in V and all a, b in $\{0, 1, 2\}$.

Example: $(a + b) = \boxed{0} \checkmark$, $(a + b)\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{0} \end{pmatrix} \checkmark$.

$a\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix} \checkmark$, $b\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix} \checkmark$
 , $a\mathbf{v} + b\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{0} \end{pmatrix} \checkmark$.

S4. $a(b\mathbf{v}) = (ab)\mathbf{v}$ for all $\mathbf{v}$ in V and all a, b in $\{0, 1, 2\}$.

Example: $b\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix} \checkmark$, $a(b\mathbf{v}) = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix} \checkmark$.

$ab = \boxed{2} \checkmark$, $(ab)\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix} \checkmark$.

S5. $1\mathbf{v} = \mathbf{v}$ for all $\mathbf{v}$ in V .

Example: $1\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix} \checkmark$.

Gap 0

Your answer is correct. You were awarded 1 mark.

Gap 1

Your answer is correct. You were awarded 1 mark.

Gap 2

Your answer is correct. You were awarded 1 mark.

Gap 3

Your answer is correct. You were awarded 1 mark.

Gap 4

Your answer is correct. You were awarded 1 mark. ✓

Gap 5

Your answer is correct. You were awarded 1 mark. ✓

Gap 6

Your answer is correct. You were awarded 1 mark. ✓

Gap 7

Your answer is correct. You were awarded 1 mark. ✓

Gap 8

Your answer is correct. You were awarded 1 mark. ✓

Gap 9

Your answer is correct. You were awarded 1 mark. ✓

Gap 10

Your answer is correct. You were awarded 1 mark. ✓

Gap 11

Your answer is correct. You were awarded 1 mark. ✓

Gap 12

Your answer is correct. You were awarded 1 mark. ✓

Gap 13

Your answer is correct. You were awarded 1 mark. ✓

Gap 14

Your answer is correct. You were awarded 1 mark. ✓

Gap 15

Your answer is correct. You were awarded 1 mark. ✓

You scored **16** marks for this part.

Score: 16/16 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).

The following two matrices are related by a sequence of elementary row operations:

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 & 1 \\ 1 & 2 & -2 & 3 & 0 \\ 2 & 4 & 1 & 1 & -2 \\ 1 & 2 & -2 & 3 & -3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$S = \{1 + x + 2x^2 + x^3, 2 + 2x + 4x^2 + 2x^3, 1 - 2x + x^2 - 2x^3\}$$

Decide whether or not S is linearly independent. If **not**, write the **second** polynomial as a linear combination of the others. If it *is* linearly independent, write 0 for both coefficients.

$$\underbrace{2 + 2x + 4x^2 + 2x^3}_{\text{polynomial 2}} = \boxed{2} \checkmark \underbrace{(1 + x + 2x^2 + x^3)}_{\text{polynomial 1}} + \boxed{0} \checkmark$$

$(1 - 2x + x^2 - 2x^3)$
polynomial 3

Your coefficients are correct. You were awarded **2** marks. ✓

You scored **2** marks for this part.

Score: 2/2 ✓

b)

(i) Find a subset of S that is a basis for W . (Select the polynomials to include in the subset)

☒ $1 + x + 2x^2 + x^3$

☐ $2 + 2x + 4x^2 + 2x^3$

☒ $1 - 2x + x^2 - 2x^3$

(ii) Hence find the dimension of W :

Correct. Dimension is 2 and polynomial 3 along with either 1 or 2 can make a basis. You were awarded **3** marks. ✓

You scored **3** marks for this part.

Score: 3/3 ✓

c)

Find a basis for the **row space** of A . Enter your basis as a matrix comprised of your basis row vectors. You can increase the number of rows as required.

Rows: Columns:

1	2	0	1	0
0	0	1	-1	0
0	0	0	0	1

Basis given for the row space of A is correct. You were awarded **1** mark. ✓

You scored **1** mark for this part.

Score: 1/1 ✓

Question 1

The following two matrices are related by a sequence of elementary row operations:

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 3 & -1 & 1 & 1 & 3 \\ 2 & 1 & -2 & -1 & 7 \\ 2 & -2 & -3 & 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \end{pmatrix}$$

Select a subset of matrices from the following which can be used to give a basis that spans $\mathbf{M}_{2,2}(\mathbb{R})$?

- ☒ $\begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix}$
☒ $\begin{pmatrix} 1 & -1 \\ 1 & -2 \end{pmatrix}$
☒ $\begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix}$
☒ $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$
- ☐ $\begin{pmatrix} 1 & 3 \\ 7 & 0 \end{pmatrix}$
☐ Not possible

Correct. The matrices selected will form a basis that spans the set of real 2x2 matrices. You were awarded **4** marks. ✓

You scored **4** marks for this part.

Score: 4/4 ✓